

FindLostPuppy — Technical Architecture, System Context & Invariants Dossier

1. System Overview & Core Mission

FindLostPuppy is a community-driven, real-time emergency pet recovery and tracking platform built for Indian pet parents and animal rescue volunteers (with optimized administrative datasets for Andhra Pradesh, Telangana, and nationwide).

The platform pairs **zero-leak personal privacy** (masking owner contact details, hiding exact home coordinates) with **rapid crowd-sourced reporting** (instant camera photo capture, GPS geocoding, and filtered sighting feeds).

2. Technology Stack & Connectors

| Layer / Connector | Technologies & Frameworks | Architecture Boundaries & Role | | :--- | :--- | :--- | |
Frontend Core | React 18, TypeScript, Vite, CSS Modules / Vanilla CSS | Single Page Application (SPA). No TailwindCSS or heavy component libraries. Custom responsive design system (320px to 1024px+). |
| **Mobile Runtime** | Capacitor 6, Android Native (Gradle 8.7+, AGP 8.3+, Java 17) | Wraps web app in Android WebView. Native dark splash screen with transparent icon. Backup disabled (`allowBackup="false"`). | |
Authentication | Firebase Authentication (Google Sign-In) | Google OAuth sign-in. Retains active session in local storage with synthetic fallback for offline/test environments. | |
Database | Google Cloud Firestore | Collections: `pets`, `sightings`, `missing_reports`, `safe_reports`, `owner_profiles`, `user_feedback`. Deny-by-default security rules. | |
Public Media | Cloudinary REST API (Unsigned Preset) | Used strictly for public missing-pet photos and sighting uploads under sanitized public alert folder. Pre-compressed before upload. | |
Private Media | Firebase Storage | Private owner profile pictures and private pet media. Restricted strictly to owner/admin access. | |
Location & Geocoding | W3C Geolocation API, OpenStreetMap / Local Census | Administrative hierarchy: State → District → Mandal/Municipality → Village/Locality. | |
Native Device Plugins | `@capacitor/camera`, `@capacitor/geolocation`, `@capacitor/app` | Native camera with gallery fallback; high-accuracy GPS with timeout protection; native Android back button handling. |

3. Security, Privacy & Compliance Invariants

CRITICAL INVARIANTS — STRICTLY ENFORCED:

1. **Zero-Leak PII Masking:** - Public owner contact details must ALWAYS remain masked: - Phone: `+91 86.....48` - Email: `j....5@gmail.com` 2. **No Exact Home GPS Coordinates or Addresses:** - Exact home

coordinates, street names, and door numbers must NEVER be exposed publicly. - Public missing-pet listings and sightings display only State, District, Mandal, and Village/Locality. 3. **Firestore Deny-by-Default:** - Pet documents and `safe_reports` are readable ONLY by their authenticated owner or an administrator. - Signed-in community users can ONLY query sanitized `missing_reports` and sanitized `sightings`. 4. **Cloudinary Unsigned Upload Constraints:** - Unsigned Cloudinary uploads are restricted strictly to public missing-pet and sighting images under the designated public folder. - Profile photos and private pet media must NEVER use unsigned Cloudinary presets. - Cloudinary API secrets must never be bundled into the web or Android client. 5. **Android Application Backup Disabled:** - In `AndroidManifest.xml`, `android:allowBackup="false"` is mandatory because local WebView storage contains authenticated profile and pet data. 6. **Strict Indian Phone Number Validation:** - Phone fields must pass `validateIndianPhoneNumber`: exactly 10 digits beginning with 6, 7, 8, or 9. 7. **Client-Side Photo Compression:** - Every photo must be compressed through `compressImage` prior to local persistence or network upload to avoid memory exhaustion and bandwidth bloat. 8. **State Mutual Exclusivity:** - A pet can NEVER be `LOST` and `SAFE` at the same time. - Selecting "Pet is Not Safe (Missing)" triggers the emergency broadcast modal; the `LOST` state is only committed after successful form submission. 9. **Sonu Memorial Invariant:** - Preserve Sonu (`#178885000505`) as `/src/assets/sonu.jpg`.

4. Component Hierarchy & Navigation Flow

```

` [App Launch] ↓ [Animated Native / React Splash] (Dark background, transparent icon) ↓
[Authentication / Login Page] (Google Sign-In) ↓ ─ YES → [Inauguration & Gratitude
Presentation] → [Owner Profile Form] ─ NO → [Dashboard Page] (Inauguration available via
footer) ↑ [Location Selector] (State → District → Mandal → Village) ↓ [Pet Choice / Details
Page] ─ Existing Pet → "You already had a pet: " ─ New Pet → "Pet Registered" (Fill Form)
─ No Pet → "Skip to Dashboard" → [Dashboard Page] `

```

Component Rules

- **Owner Profile:** Saved Owner Details rows are read-only and show no inline orange pencil edit affordances. `Modify Details / Photo` is the single entry point for editing.
- **Pet Choice / Onboarding:**
 - Heading is `Pet Registered` when a pet exists. - Existing pet choice reads `You already had a pet:` . - Exactly one deletion control: top `Remove Pet` button. - Prominent `Skip to Dashboard` button enables pet-free users to participate in community sightings immediately.

5. Location Hierarchy & Geocoding Rules

FindLostPuppy enforces a 4-tier geographic administrative division: 1. **State:** Top-level selection (e.g., Andhra Pradesh, Telangana). Changing State resets District, Mandal, and Village. 2. **District:** Dependent on State. Changing District resets Mandal and Village. 3. **Mandal / Municipality:** Sub-district administrative unit. Distinct from Village/Locality. 4. **Village / Home Base:** - **Distinct Village Invariant:** Village/Home Base must contain the specific village or locality and must NEVER repeat the selected mandal. - Village selectors

filter out any option identical to the selected mandal. - If autodetect cannot resolve a distinct village, Village is left blank for manual user selection.

Capture Pet Location Invariants

- State, District, Mandal, and Village initialize as blank, even if the owner profile has a saved home address.
- Cascading filters: changing parent clears all child fields (no auto-selection).
- Dropdowns retain green pet-present indicators (●) showing missing pet counts at that administrative tier.

6. Dashboard Pet Categories & Modal Rules

The Dashboard revolves around three primary category cards:

- **Sighted Missing Pets:** Sightings reported by community members with photos and geolocations.
- **Pets at Home:** Verified safe pets registered to their owners.
- **Pets Missing:** Pets broadcasted as missing, awaiting community recovery.

Modal Invariants

- **Bounded Internally Scrollable Modals:** Selecting a category opens an internally scrollable modal; it never appends long lists beneath cards on the main page.
- **Strict Single-Modal Rule:** Only one modal is permitted on screen at any time. Opening a pet detail or sighting detail modal immediately hides the category list modal.
- **Context Restoration:** Closing a detail modal immediately restores the previously active category list modal at its exact scroll context.
- **Grouped Sightings:** Multiple sightings for the same missing pet are consolidated behind **View sightings (n)** action. Sighting detail dialog displays numbered tabs (**Sighting 1** , **Sighting 2** , etc.).
- **Safe Pet Owner-Only Privacy:** Safe-pet **View Details** is available ONLY to the signed-in owner. Other users cannot view safe pet profiles. Safe-pet dialogs are read-only and never show "Mark Missing" or "Edit" buttons.

7. Emergency Alert & Roaming Pet Capture System

Quick Roaming Pet Capture (On Dashboard)

- **Triggers:** Header button `#dashboard-quick-capture-btn` and Floating Action Button (FAB) `#dashboard-floating-capture-fab`.
- **Instant Geolocation & Camera:** User snaps photo; app auto-captures current GPS coordinates.
- **Automatic Designation:**
 - Pet ID: `UNKNOWN_ROAMING_PET` - Pet Name: `"Unknown (Roaming Pet)"` - Status: `LOST`
- **Instant Discovery:** Sighting is immediately submitted into the "Pets Missing" repository so owners can filter and locate their lost pets.

Cascading Location Filter Bar

- Dropdown filter bar (State → District → Mandal → Village) inside the "Pets Missing" modal lets users quickly find lost and roaming reports near their neighborhood.

App Suggestion & Feedback Widget Modal

- Never pops up automatically on first dashboard load.
- Accessible via a button at the bottom of the dashboard.
- Clean layout: App Rating & Feedback header, interactive 5-star rating, feedback comment box, centered submit button.

8. Developer & Release Engineering Cheat Sheet

Terminal Commands

```
` bash
```

1. Start local dev server (Port 5173)

```
npm run dev
```

2. Stop local dev server

```
kill -9 $(lsof -ti:5173) 2>/dev/null || pkill -f "vite" `
```

Pre-Release Verification Chain

```
` bash
```

Run before any release, commit, or PR:

```
npm run lint # Must have 0 errors npm run build # TypeScript & Vite production build npm run  
sync:android # Sync web assets to Capacitor Android npm run build:android # Gradle build  
release AAB / APK node scripts/test_responsive.mjs # Responsive verification (320px, 390px,  
768px, 1024px) `
```

Android Release Artifact Requirements

Copy release bundle from Gradle output to Desktop under both filenames: 1.

/Users/jayakrishna/Desktop/findlostpuppy-release.aab 2.

/Users/jayakrishna/Desktop/findlostpuppy-v{versionName}-code{versionCode}-release.aab

9. Prompt Template for Claude AI Context

```
`
```

```
` text You are an expert full-stack engineer pair-programming on the FindLostPuppy project.  
Here is the authoritative system context and non-negotiable invariants:
```

```
1. REPOSITORY & WORKFLOW RULES:
```

- Always work locally on the 'fixes' branch. Never commit or push directly to 'main'.

- Never modify production without user approval.
- Local dev server runs on `http://localhost:5173`.
- After any UI edit, run: `npm run lint`, `npm run build`, `npm run sync:android`, and `node scripts/test_responsive.mjs`.
- Ensure layouts work flawlessly from 320px, 390px, to 1024px+ viewports without horizontal scroll.

2. SECURITY & PRIVACY:

- Strict PII masking: Mask owner phone numbers (+91 86*****48) and emails (j***5@gmail.com).
- Never expose exact home GPS coordinates or street addresses in public listings.
- Firestore is deny-by-default. Private pet docs and SAFE reports are owner/admin only. Community users read only sanitized LOST reports and sightings.
- Unsigned Cloudinary is restricted to sanitized public missing-report photos. Private photos use Firebase Storage.
- Android backup is disabled (`android:allowBackup="false"`).
- Indian phone number validation: exactly 10 digits beginning with 6, 7, 8, or 9.
- Always compress photos via `compressImage` before saving or uploading.
- Preserve Sonu (#1788885000505) in `/src/assets/sonu.jpg`.

3. LOCATION INVARIANTS:

- 4 tiers: State → District → Mandal/Municipality → Village/Locality.
- Village must be distinct from Mandal (never duplicate the mandal name).
- Capture Pet page always initializes with blank location selectors.
- Cascading dropdowns: changing a parent clears all child fields (no auto-selection).
- Dropdowns retain green pet-present indicators (●) showing counts.

4. NAVIGATION & DASHBOARD:

- Onboarding order: Auth → Inauguration (first-time only) → Owner Profile → Location → Pet Choice/Details → Dashboard.
- Dashboard has 3 category cards: 'Sighted Missing Pets', 'Pets at Home', 'Pets Missing'.
- Selecting a card opens an internally scrollable modal. Single-modal visibility rule: opening a detail modal hides the category modal; closing it restores the category modal.
- Sighted reports for the same pet are grouped behind 'View sightings (n)' with numbered tabs (Sighting 1, 2...).
- Quick capture on Dashboard logs roaming pets as 'Unknown (Roaming Pet)' with ID `UNKNOWN_ROAMING_PET` directly into 'Pets Missing'.